

1 (a) force per unit mass (ratio idea essential) B1 [1]

(b) graph: correct curvature M1
from $(R, 1.0 g_S)$ & at least one other correct point A1 [2]

(c) (i) fields of Earth and Moon are in opposite directions M1
either resultant field found by subtraction of the field strength
or any other sensible comment A1
so there is a point where it is zero A0 [2]
(allow $F_E = -F_M$ for 2 marks)

(ii) $GM_E / x^2 = GM_M / (D - x)^2$ C1
 $(6.0 \times 10^{24}) / (7.4 \times 10^{22}) = x^2 / (60R_E - x)^2$ C1
 $x = 54 R_E$ A1 [3]

(iii) graph: $g = 0$ at least $\frac{2}{3}$ distance to Moon B1
 g_E and g_M in opposite directions M1
correct curvature (by eye) and $g_E > g_M$ at surface A1 [3]

2 (a) (i) no forces (of attraction or repulsion) between atoms / molecules / particles B1 [1]